

Esequiel Linares

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EDUCATION

Purdue University

Bachelor of Science in Computer Science

West Lafayette, IN

Expected Graduation, May 2027

EXPERIENCE

Undergraduate Researcher – Multimodal Generative AI

September 2025 – Present

Purdue University

West Lafayette, IN

- Led research on Duet Dance dataset (10.3 hrs, 15 genres, 30 dancers, 10K+ annotations, 4K sequences), achieving up to 6× accuracy gains in duet generation.
- Built PyTorch pipelines for Text-to-Duet and Dance Accompaniment, improving models by 60–500%.
- Analyzed VR/XR and multimodal embeddings, identifying 5 use cases and contributing to a research paper.

Backend Developer – Code for Good Hackathon

November 2025

JPMorgan Chase & Co.

Columbus, OH

- Developed 10+ Flask RESTful endpoints using SQLAlchemy ORM, implementing robust CRUD operations, error handling, and CORS configuration for frontend integration.
- Built automated CSV ingestion pipeline transforming raw data into structured database records, enabling real-time dashboard updates through backend-frontend API integration.
- Authored 56 commits (47% of repository), leading backend architecture and deploying to AWS EC2 to achieve top-3 placement in 24 hours.

Capital One First Gen Focus

May 2025 – July 2025

Participant (Top 10% Selected)

Chicago, IL (Virtual)

- Chosen from 1,000+ applicants for an 8-week program focused on empowering first-generation students.
- Executed 10+ hours of coding workshops on LeetCode prep and software engineering career navigation.
- Advanced backend and ML skills through Capital One SWE mentorship, building 3 projects including a multilayer perceptron that improved name-generation accuracy by 25% and diversity by 40%.

PROJECTS

Reachy Mini (HCI) | C++, Hugging Face, PyTorch, WebRTC, RunPod (Cloud GPUs), CUDA August 2025 – Present

- Architected a humanoid-inspired robotic assistant that fuses speech, vision, and decision-making through a low-latency cloud-edge pipeline combining Raspberry Pi 5 with GPU servers.
- Integrated Hugging Face Transformers for multimodal emotion and intent recognition, elevating interaction accuracy by 40% and enabling adaptive, context-aware dialogue.
- Engineered WebRTC streaming and optimized C++ drivers for Pi 5 speaker/camera modules, reducing I/O delays by 25% and sustaining sub-100ms end-to-end response times.

AirCursor (Computer Vision) | Google Mediapipe, OpenCV, NumPy, Google Colab August 2024 – November 2024

- Built a CNN on 1,440 video frames, reaching 95% accuracy in hand-gesture recognition.
- Accelerated data processing, cutting training time to 5 minutes and enabling rapid gesture creation.
- Implemented a low-latency cursor control system with PyAutoGUI and OpenCV, enhancing real-time interaction.

Shop22 (E-commerce) | TypeScript, Stripe API, Sass, Payload CMS, Next 14, Docker June 2024 – December 2024

- Developed an e-commerce platform with React and Payload CMS, enhancing shopping with customized product listings.
- Implemented secure payment processing with Stripe API, optimizing transaction flow and customer trust.
- Configured Docker for seamless deployment, ensuring scalable and maintainable application architecture.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, JavaScript, TypeScript, SQL, HTML, Bash

Web & Styling Tools: CSS, Sass, Tailwind CSS

Frameworks & Libraries: Flask, React, Vite, Node.js, Next.js, Spring Boot, Java Swing, JavaFX, React Router, PyTorch, Hugging Face, NumPy, Pandas, Matplotlib, Mediapipe, PyAutoGUI, Apache Commons, CUDA, JUnit, pytest

Databases: PostgreSQL, SQLite

Developer Tools & Platforms: Git, Docker, AWS (EC2), Google Cloud Platform, RunPod, VS Code, Cursor AI, Visual Studio, CLion, IntelliJ, PyCharm, Linux/Unix